

Samaksh Judson

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EDUCATION

Carnegie Mellon University

Master of Science in Mechanical Engineering (CDM)

Pittsburgh, PA

Dec 2024

Coursework - Deep Reinforcement Learning for Control, Path Planning and Decision Making, Visual Learning and Recognition, Optimal Control and Reinforcement Learning, Introduction to Deep Learning, Engineering Computation

Birla Institute of Technology and Science, Pilani

Bachelor of Engineering in Mechanical Engineering

Pilani, India

June 2023

SKILLS

Programming Languages: Python (NumPy, PyTorch, Pandas, OpenCV, Matplotlib), C, C++, Julia

Robotics Frameworks: ROS1, ROS2, Gazebo, RViz, Isaac Sim, Mujoco

Development Tools: Git, Docker, Kubernetes, API WebSocket integration, Cloud deployment (AWS/GCP), Bash scripting

WORK EXPERIENCE

Onward Robotics

Robot Software Engineer II, Planning & Perception

Pittsburgh, PA

Dec 2025 – Present

- Led greenfield development of a production-grade AMR autonomy stack in ROS2, implementing sensor drivers (camera, motor, LiDAR), localization using graph-based optimization, and trajectory planning and control.
- Benchmarked and tuned global and local planners, optimizing world representation, costmaps, and control-loop scheduling, reducing mission traversal time from **88s to 38s (57% improvement)** under identical constraints.
- Architected the autonomy integration layer connecting perception, state estimation, and motion planning modules, designing fault-handling and recovery logic ensure deterministic operation under sensor or actuator failures.
- Implemented temporal voxel-based sensor fusion within the world model and integrated SICK safety fields as a hardware-redundant safety layer independent of the autonomy application stack.
- Profiled and optimized latency-critical C++ execution paths in the planning pipeline, reducing planner/controller compute latency by **26%** and improving real-time stability.

Robot Software Engineer I, Planning & Perception

Mar 2025 – Dec 2025

- Developed CUDA-accelerated modules for perception and motion planning, increasing throughput by **65%** across multi-sensor data streams.
- Implemented a RANSAC-based visual-inertial alignment module for pose refinement, improving localization stability and reducing drift in dynamic warehouse environments.
- Built infrastructure for telemetry logging and REST/WebSocket interfaces to support remote monitoring and debugging of on-vehicle behavior.

fpprime AI | Python, C++ Intern

Pittsburgh, PA

Feb 2025 - March 2025

- Fine-tuned vision, language, and audio foundation models to infer high-level contextual cues for a productivity-monitoring state machine, achieving **81%** state transition accuracy under real-world conditions.
- Developed a multimodal fusion network with modal dropout to maintain performance under sensor occlusion or signal loss, boosting task recognition accuracy by **22%** over unimodal baselines in dynamic environments.

Kantor Lab | ROS1, ROS2, Docker, PyTorch, C++, Arduino Graduate Research Assistant

Pittsburgh, PA

May 2024 – Aug 2024

- Developed distributed, asynchronous pipelines for real-time 3D point cloud modeling and occluded fruit detection, integrating YOLOv4 for segmentation (mAP **91.3%**), RAFT_stereo for depth estimation (EPE **0.56 px**), and DeepSORT for tracking (IDF1 **87.6%**).
- Implemented EKF-based visual-inertial odometry for real-time pose estimation, reducing drift and improving localization accuracy, achieving a **12%** reduction in pose RMSE compared to baseline methods.

PROJECTS

Imperative Path Planning | Python, ROS, Isaac Sim, AWS

- Developed a novel unsupervised path planning policy for autonomous driving in unstructured environments.
- Implemented a custom-trained variational autoencoder (VAE) with a frozen decoder architecture, using RMSE loss to validate improvements in the quality of encoder embeddings for better generalization.
- Created an LSTM-based architecture for waypoint generation, enabling smoother trajectory predictions and improved obstacle avoidance.
- Achieved **SOTA** zero-shot performance in obstacle avoidance and waypoint generation, outperforming conventional unsupervised learning techniques as measured by the Success weighted by Path Length (SPL) metric.